

Dear Professor Marc & Selection Team,

I am applying for the EngD on developing an **interactive reverse logistics map** for circular construction. I find this topic very interesting because a map can bring material information, logistics and decisions into one place, and help good materials get a second life.

During my master's studies, my thesis, **Agile Project Management in the Construction Industry**, looked at how Agile can improve teamwork, risk management, decision-making and the use of resources on construction projects. After finishing it, I saw a clear limit: better project delivery and less construction waste only cover one part of sustainability. This led me to a bigger question: what happens to materials when buildings reach the end of their life?

When I looked into circular construction and reverse logistics, and read earlier research including Professor Voordijk's work on recovering building elements for reuse, that question became clearer. Reuse is not only a technical issue. It also needs good coordination from the donor building to the target building. Information about availability, quality, ownership, location, transport and reuse options is often missing or not shared. So new materials stay the safer choice. For the map, I see three practical questions: How can people see which materials are available and in what condition? How can we match donor supply with target demand when timing, quality and cost are uncertain? How can a map help users compare reuse options on cost, CO₂ and circularity, without becoming a tool that nobody uses? These are the questions I want to work on.

My experience fits this well. At Aug.e I worked on digital twins for smart buildings, using a digital model to understand and improve how a real building works. At Hive CPQ I managed large product datasets and turned complex information into simple digital tools for non-technical users. At iostiring I worked with a construction partner on a circular construction hypothesis: how Agile project management can bring circular practices into real projects. That work stayed at the hypothesis stage, so a (de)construction pilot with partners such as Dura Vermeer and TNO is the natural next step for me. Together with my civil and environmental engineering background and GIS skills, this is what I can bring to the project. I recently took a career break to care for my newborns, and during that time I kept learning in this field. I am ready to move to the Netherlands for this EngD.

The EngD at the University of Twente is exciting for me because it combines strong study with a real design project in industry. I want to grow in reverse-logistics modelling, GIS for material matching, and design-science research: define use cases with partners, test the map in a pilot, and improve it with the people who will use it. Twente is a great place for this because research and industry work closely together.

Thank you for reading my application. I would be happy to talk about how I can contribute to the project.

Yours sincerely,

Pooja Awasare